Problem Set for 12-May and 15-May

The ambient field is $\mathbb{R}$.

Exercise 1 Determine whether the following are inner product spaces (简要回答即可).

1. V is the vector space continuous functions over $\mathbb{R}$, and the pairing is
$$(f,g) \mapsto \int_0^1 f(x)g(x)\,\mathrm{d}\,x.$$
2. $V = \mathbb{R}[x]$, and the pairing is
$$(f,g) \mapsto \int_{-\infty}^{\infty} f(x)g(x)e^{-x^2}\,\mathrm{d}\,x.$$
3. $V = \mathbb{R}^{n\times n}$, and the pairing is
$$(A,B) \mapsto \det(A^T \cdot B).$$
4. (**optional**) By axiom of choice, any $\mathbb{R}$-vector space can be endowed with an inner product structure.

Exercise 2 Show that every real matrix is similar to a block diagonal matrix, wherein the candidates of the blocks are

1. the Jordan form of a real eigenvalue;
2. the block of the form
$$\begin{bmatrix} H_{r,\theta} & Q & & & \\ & H_{r,\theta} & Q & & \\ & & \ddots & \ddots & \\ & & & H_{r,\theta} & Q \\ & & & & H_{r,\theta} \end{bmatrix},$$
where
   1. $H_{r,\theta} = \begin{bmatrix} r\cos\theta & -r\sin\theta \\ r\sin\theta & r\cos\theta \end{bmatrix}$, and
   2. $Q = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.

> ⓘ **Note**
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> When proving with Jordan form, show that $A \sim B$ over $\mathbb{R}$ if and only if $A \sim B$ over $\mathbb{C}$.

Exercise 2.5 Show that every real matrix is a product of two real symmetric matrices.

Exercise 3 Let $(V, (-,-))$ be a finite dimensional real inner product space. For unit vector v_0 (i.e., $(v_0, v_0) = 1$), we define the reflection

$$\varphi : V \to V, \quad x \mapsto x - 2(x, v_0) \cdot v_0.$$

Show that for isometric transform ψ (i.e., $(\psi(u), \psi(v)) = (u, v)$), 1 is an eigenvalue for φ or $\psi \circ \varphi$. Overall, $1 \in \sigma(\varphi) \cup \sigma(\varphi \circ \psi)$.

> **Note**
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> Matrix version of the problem?

Exercise 4 Set $V := \mathbb{R}[x]$ and $V_0 := \{f \in \mathbb{R}[x] \mid f(0) = f(1)\}$.

1. Prove that $V \times V \to \mathbb{R}, \quad (f, g) \mapsto \int_0^1 f(x)g(x)\,\mathrm{d}\,x$ is an inner product.
2. Set $\mathscr{D} : V_0 \to V, \quad f(x) \mapsto f'(x)$. Find $\dim\ker(\mathscr{D})$ and $\dim\operatorname{coker}(\mathscr{D})$.
3. Define the inner product restricted on the subspace

$$(\cdot,\cdot)_0 : V_0 \times V_0 \to \mathbb{R}, \quad (f, g) \mapsto \int_0^1 f(x)g(x)\,\mathrm{d}\,x.$$

   Is there any linear map $\mathscr{D}^* : V \to V_0$ such that for any $h \in V_0$ and $g \in V$,

$$(\mathscr{D}^* g, h)_0 = (g, \mathscr{D}h)?$$

> **Tip**
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> 去年期末考.

Exercise 5 Let $(V, (,))$ be an inner product space, where V is not necessary finite dimensional.

- Say $\varphi^\star$ is the adjoint of $\varphi \in \mathrm{Hom}_{\mathbb{R}}(V, V)$, provided $(\varphi^\star(x), y) = (x, \varphi(y))$ for arbitrary $x, y \in V$.
- Let U be a subspace of V. Set

$$U^\perp := \{v \in V \mid (u, v) = 0, \forall u \in U\} = \bigcap_{u \in U} \ker((u, -))$$

Now we assume that φ is invertible, and $\varphi^\star$ exists.

1. Show that $\varphi^\star$ is injective, and $(\mathrm{im}(\varphi^\star))^\perp = 0$.
2. Show that $\varphi^\star$ is surjective $\implies (\varphi^{-1})^\star$ exists and $(\varphi^{-1})^\star = (\varphi^\star)^{-1}$.
3. Show that $(\varphi^{-1})^\star$ exists $\implies \varphi^\star$ is invertible and $(\varphi^{-1})^\star = (\varphi^\star)^{-1}$.
4. Let $V = \mathbb{R}[x]$ and $(f, g) := \sum_{n \geq 0} f_i g_i$, where $h = \sum h_i \cdot x^i$. Show that

1. $(V, (,))$ is an inner product space;
2. $L : f \mapsto \frac{f - f(0)}{x}$ is the linear map of moving left. Show that $(\mathrm{id} + L)$ is invertible;
3. Show that $\varphi := (\mathrm{id} + L)^{-1}$ has no adjoint $\varphi^\star$.

Exercise 6 (**Optional, very interesting**). Find the dimension and inertia of the following symmetric bilinear form, and verify the invariant group action.

1. $V = \{X \in \mathbb{R}^{2\times 2} \mid \mathrm{tr}(X) = 0\}$, and the pairing is $(A, B) \mapsto \mathrm{tr}(A \cdot B)$. Moreover, for $g \in \mathrm{GL}_2(\mathbb{R})$,the pairing is invariant under $X \mapsto g \cdot X \cdot g^{-1}$.

> ***Step 0*** $\dim V = 3$.
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> ***Step 1*** *The pairing is linear (trivial) and symmetric, since* $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.
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> ***Step 2*** *Let* $(A_1, A_2, A_3) := (E_{1,2}, E_{2,1}, E_{1,1} - E_{2,2})$ *be a basis of* V*, the associated Gram matrix is the* 3×3 *symmetric matrix* $(\mathrm{tr}(A_i A_j))_{3\times 3}$ *is* $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. *The inertia is* $(+ + -)$.
>
> ***Step 3*** *Since* $\mathrm{tr}((gXg^{-1}) \cdot (gYg^{-1})) = \mathrm{tr}(XY)$, *pairing is invariant under such group action.*

2. $V = \mathbb{R}^{2\times 2}$, and the pairing is $(A, B) \mapsto \mathrm{tr}(\mathrm{adj}(A) \cdot B)$. Moreover, for $(g, h) \in \mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$, the pairing is invariant under $X \mapsto g \cdot X \cdot h^{-1}$.
3. $V = \left\{ \begin{pmatrix} a & ib \\ ic & \overline{a} \end{pmatrix} \mid a \in \mathbb{C}; b, c \in \mathbb{R} \right\}$, the pairing is $(X, Y) \mapsto \mathrm{Re}(\mathrm{tr}(X \cdot \overline{Y}))$. Moreover, for $g \in \mathrm{GL}_2(\mathbb{C})$, the pairing is invariant under $X \mapsto g \cdot X \cdot (\overline{g})^{-1}$.
4. (**Optional**, if you know tensor product). Let $U \wedge U$ be the quotient space of $U \otimes U$ by the relation $(a \otimes b + b \otimes a = 0)$. In particular, $\dim(U \wedge U) = \binom{\dim U}{2}$. Let $V = \mathbb{R}^4 \wedge \mathbb{R}^4$ be the 6-dimensional vector space with basis $\{e_i \wedge e_j\}_{1 \le i < j \le 4}$, and the pairing is $(\lambda, \mu) \mapsto \lambda \wedge \mu$. Here $\bigwedge_{i=1}^4 \mathbb{R}^4$ is one-dimensional with basis $\{e_1 \wedge e_2 \wedge e_3 \wedge e_4\}$. Moreover, for $g \in \mathrm{SL}_4(\mathbb{R})$, the pairing is invariant under $\mu \mapsto g \cdot \mu$. Here $g \cdot (\sum c_{i,j} e_i \wedge e_j) := \sum c_{i,j} (g \cdot e_i) \wedge (g \cdot e_j)$.

The above calculation provides the elementary construction of low dimensional spin groups, that is, the double covering of special orthogonal groups. One learns from these construction that

1. $\mathrm{SL}_2(\mathbb{R})/\{\pm I\} \simeq \mathrm{SO}(2, 1)$, and $\mathrm{SL}_2(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}_3(\mathbb{C})$;
2. $(\mathrm{SL}_2(\mathbb{R}))^2/\{\pm(I, I)\} \simeq \mathrm{SO}(2, 2)$, and $(\mathrm{SL}_2(\mathbb{C}))^2/\{\pm(I, I)\} \simeq \mathrm{SO}_4(\mathbb{C})$;
3. $\mathrm{SL}_2(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}(3, 1)$;

4. $\mathrm{SL}_4(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}_6(\mathbb{C})$, and $\mathrm{SL}_4(\mathbb{R}) \simeq \mathrm{SO}(3,3)$.

As corollaries, $\mathrm{SO}_4(\mathbb{C})$ is a double cover of $\mathrm{SO}_3(\mathbb{C})$, and the group isomorphism $\mathrm{SO}(3,1) \simeq \mathrm{SO}_3(\mathbb{C})$ is a famous coincidence in quantum physics.